ITAD329

Mobile computing

CIA - I

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① Given:

Total area = 1200 km^2

Cell radius = 1.8 km

Spectrum Available = 48 MHz

Full-duplex channel = 200 kHz

Reuse cluster sizes N = 7

N = 4

Reduction = 15%, Minimum reuse ratio Q = 4.5

calculate the no. of cells, total available channels, theoretical and effective channels per cell, and co-channel reuse distance for both configuration.

Area = $2.958 \times (1.8)^2$

$= 2.958 \times 3.24$

$= 8.418$

Number of cells = $\frac{1200}{8.418}$

= 143 cells

N = 4

$= R\sqrt{3N}$

$= 1.8\sqrt{3 \times 4}$

$= 1.8\sqrt{12}$

$= 306.24$ km

Ratio $= \sqrt{3N}$

channels:

48 MHz convert into KHz

$48 \times 1000 = 48000$

$= \frac{48000}{200}$

$= 240$ KHz

channels per cell.

$N=7 \Rightarrow \frac{240}{7} = 34.29$

15 % reduction $\Rightarrow 0.85$

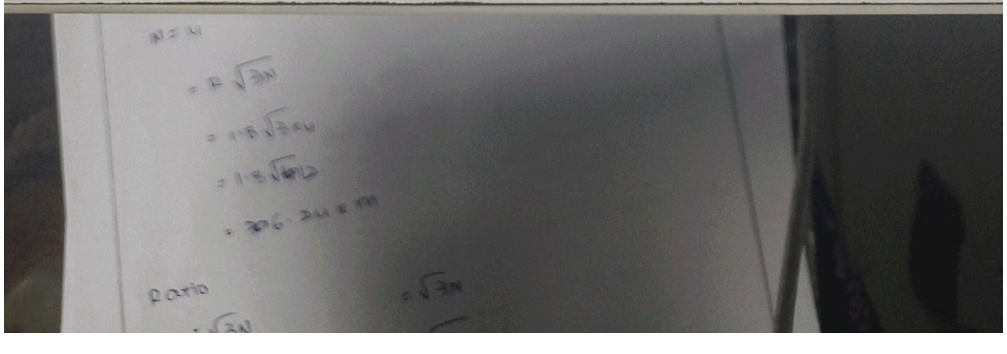
$= 34.2 \times 0.85$

$= 29.14$

$N = 4 \Rightarrow \frac{240}{4} = 60$

$\Rightarrow 60 \times 0.85$

$= 51$



Reuse distance $D = R\sqrt{3N}$

$= 1.8\sqrt{3 \times 7}$

$= 1.8\sqrt{21}$

$= 8.24$

$N = 4$

$= R\sqrt{3N}$

$= 1.8\sqrt{3 \times 4}$

$= 1.8\sqrt{12}$

$= 6.24$ km

Ratio

$= \sqrt{3N}$ $= \sqrt{3N}$

$= \sqrt{3 \times 7}$ $= \sqrt{3 \times 4}$

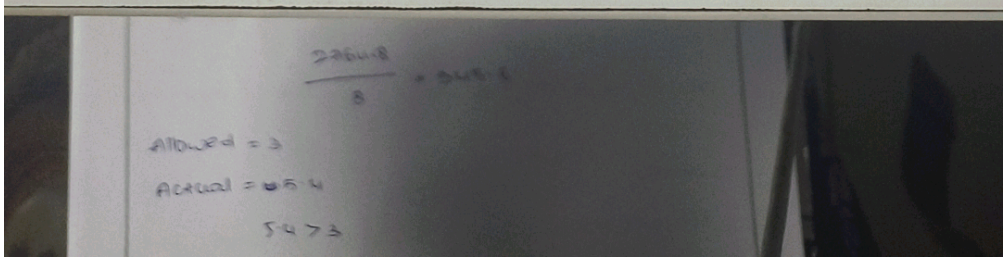
$= \sqrt{21}$ $= \sqrt{12}$

$= 4.583$ $= 3.464$

Minimum reuse ratio $Q = 4.5$

$N = 7$ is greater than reuse ratio $4.5 \rightarrow$ pass

$N = 4$ is lesser than reuse ratio $4.5 \rightarrow$ Fail



(2) Given:

BSS = 120 ms

NSS = 80 ms

For one hand off : 120 + 80

= 200

For 15 hand off ⇒ 15 × 200

= 3000 ms

Convert milliseconds to seconds

$\frac{3000}{1000}$ = 3 sec

Failed 3 sec for 400 ms

⇒ 3 × 400 = 1200 ms

⇒ 1.2 seconds

Roaming delay

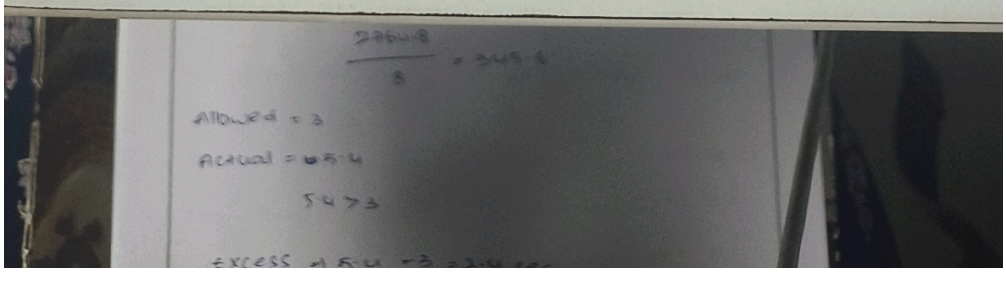
Authentication = 350 ms

Registration = 250 ms

⇒ 350 + 250 = 600 ms

⇒ 2 × 600 = 1200 ms

⇒ 1.2 sec.



$= 3 + 1.2 + 1.2$

$= 5.4$ sec

Potential patient data = 512 kbps

Data affected = Data rate x Time

$= 512 \times 5.4$

$= 2764.8$ k bits

convert into kB

$$\frac{2764.8}{8} = 345.6$$

Allowed = 3

Actual = 5.4

$5.4 > 3$

excess = $5.4 - 3 = 2.4$ sec

(3)

i) Given:

Subscribers = 1200

call per hour = 1.2

call holding time = 90 sec

Time slots TRX = 8 , 2

BTS = 12

Sectors = 3

Transceivers (TRX S) = 4

Utilization limit = 80%.

BSC single E1 link = 2.048 Mbps

Abis interface = 16 Kbps

⇒ Transceivers x Time slot (TRX)

⇒ 4 x 8 = 32

Traffic channels = 32 - 2

= 30

channels

For BTS = 30 x 3 = 90

For BSC = 90 x 12 = 1080

Traffic:

$$A = \frac{\text{subscribers} \times \text{call per hours} \times \text{call holding}}{3600}$$

$$= \frac{1200 \times 1.2 \times 90}{3600}$$

= 36 Erlang

$\Rightarrow \frac{\text{offered traffic}}{\text{available channey}} \times 100$

$= \frac{36}{30} \times 100$

$= 120\%$.

Abis interface $\Rightarrow 90 \times 16$

$\Rightarrow 1440$ kbps

$= 1.44$ Mbps

Utilization limit $= 80\%$.

$= \frac{36}{0.80} = 45$

Two signaling $\Rightarrow 45+2 = 47$

8 time slots

$\frac{47}{8} = 5.875$

Round off $= 6$

New traffic channel $\Rightarrow (6 \times 8) - 2$

$= 46$

$\frac{36}{46} \times 100 = 78.26$

(4) Given:

Network transfers = 240

Size of each = 18 MB

Data rate = 20 Kbps

capacity reduces = 18%.

Congestion throughput = 85%.

packet retransmission = 12%.

interruptions = 2.5 sec

UMTS transfer time = 384 Kbps.

(5) Given:

software = 120 GB

mobile devices = 2,000

GSM = 9.6 Kbps per connection

GPRS = 64 Kbps

- GPRS with 10% addition retransmission traffic

5G = 25 Mbps

- 5G provides 4% protocol

∴ 40% of devices can transfer data over 5G

Assuming 1GB = 1000MB

1 MB = 8 megabits

$\Rightarrow 120 \times 1000 \Rightarrow 120000$ MB

$\Rightarrow \frac{120000}{2000} = 60$ MB

convest it into megabits

$60 \times 8 = 480$

GSM transfer $= \frac{9.6}{1000} = 0.00096$

$\Rightarrow \frac{480}{0.00096} = 5000$ sec

convert into hours

$\Rightarrow \frac{5000}{3600}$

$= 13.89$ hours

GPRS transfer

$\Rightarrow 480 \times 1.10 = 528$ Mb

convest into speed

$\frac{64}{1000} = 0.064$ Mbps

$$T = \frac{528}{0.064}$$

$$= 8250$$

$$= 8250/3600$$

$$= 2.29 \text{ hours}$$

5G

$$\Rightarrow 25 \times 0.96$$

$$= 24 \text{ Mbps}$$

Transfer time

$$T = \frac{480}{24}$$

$$= 20 \text{ sec}$$

no. of batches

$$2000 \times \frac{40}{100} = 800$$

$$\text{Total} = \frac{2000}{800} = 2.5$$

$$= 3$$

$$3 \times 20 = 60 \text{ sec}$$

$$\text{percentage} = \frac{\text{GPRS} - 5G}{\text{GPRS}} \times 100 = \frac{8250 - 60}{8250} \times 100$$

$$= 99.27\%$$